Elementary Analysis

Chapter 2, Section 10

Monotone Sequences and Cauchy Sequences
Exercises 10.1–10.12, with every subpart

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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10 Monotone sequences and Cauchy sequences

An increasing sequence satisfies $s_n \le s_{n+1}$; a decreasing sequence satisfies $s_n \ge s_{n+1}$. Both definitions permit equality. A sequence is bounded if all its terms lie between two fixed real numbers.

The Cauchy condition says that, for every $\varepsilon > 0$, there is one threshold N such that

$$m > N,\ n > N \quad \Longrightarrow \quad |s_m - s_n| < \varepsilon.$$

It controls every pair of late terms. Merely making successive differences small is not enough, as Exercise 10.6 explains. The solutions first justify convergence and then identify a limit.

Conventions for monotonicity and Cauchy estimates

In Definition 10.1, "increasing" means $s_n \le s_{n+1}$ and "decreasing" means $s_n \ge s_{n+1}$. Equality is permitted in both definitions. A constant sequence therefore has both properties. Repeated application of an increasing inequality gives $s_n \le s_m$ whenever $n < m$; for a decreasing sequence it gives $s_n \ge s_m$.

Definition 10.8 says that (s_n) is Cauchy if

> for every $\varepsilon > 0$ there is an N such that every pair of integers $m, n > N$ satisfies $|s_m - s_n| < \varepsilon$.

Both indices are arbitrary after the same threshold. An estimate for consecutive terms alone is not yet the estimate for an arbitrary pair.

10.1 Exercise 10.1: Classifying six sequences

What we must show. Decide whether each sequence is increasing, decreasing, and bounded, with the weak monotonicity conventions just stated.

Part (a): the sequence $s_n = 1/n$.

For every $n \in \mathbb{N}$, the denominators n and $n+1$ are positive, and

$$s_{n+1} - s_n = \frac{1}{n+1} - \frac{1}{n} = \frac{n-(n+1)}{n(n+1)} = -\frac{1}{n(n+1)} < 0.$$

Thus $s_{n+1} < s_n$ for every n, so the sequence is decreasing. It is not increasing: for example, $s_2 = 1/2 < 1 = s_1$. Because $n \ge 1$, we also have $0 < 1/n \le 1$, so 0 and 1 are a lower and an upper bound.

Conclusion. The sequence $1/n$ is **decreasing and bounded**, and is not increasing.

Part (b), Step 1: test monotonicity with exact terms.

Here $s_n = (-1)^n/n^2$. Its first three terms are

$$s_1 = -1, \qquad s_2 = \frac{1}{4}, \qquad s_3 = -\frac{1}{9}.$$

The inequality $s_1 < s_2$ rules out a decreasing sequence, which would require $s_1 \ge s_2$. The inequality $s_2 > s_3$ rules out an increasing sequence, which would require $s_2 \le s_3$. Hence it is neither.

Part (b), Step 2: give bounds valid at every index.

If n is odd, then $s_n = -1/n^2$, so $-1 \le s_n < 0$. If n is even, then $n \ge 2$ and $s_n = 1/n^2$, so $0 < s_n \le 1/4$. Both cases give

$$-1 \le s_n \le \frac{1}{4} \qquad (n \in \mathbb{N}).$$

These bounds are attained at $n = 1$ and $n = 2$, respectively.

Conclusion. The sequence $(-1)^n/n^2$ is **bounded but neither increasing nor decreasing**.

Part (c), Step 1: check the growth of $s_n = n^5$.

For every $n \ge 1$, the binomial expansion gives

$$\begin{aligned} s_{n+1} - s_n &= (n+1)^5 - n^5, \\ &= n^5 + 5n^4 + 10n^3 + 10n^2 + 5n + 1 - n^5, \\ &= 5n^4 + 10n^3 + 10n^2 + 5n + 1 > 0. \end{aligned}$$

All five terms in the final sum are positive. Thus $s_{n+1} > s_n$ for every index, giving an increasing sequence and ruling out a decreasing one.

Part (c), Step 2: prove unboundedness above.

Since $n \ge 1$, we have $n^4 \ge 1$, so $n^5 \ge n \ge 1$. The number 1 is a lower bound. To show there is no upper bound, let $U \in \mathbb{R}$ be any proposed upper bound. By the Archimedean property, choose $n \in \mathbb{N}$ with $n > \max\{1, U\}$. Then

$$s_n = n^5 \ge n > U.$$

Thus U fails to bound the sequence above. This works for every real candidate U.

Conclusion. The sequence n^5 is **increasing and unbounded above**. It is not decreasing or bounded, although it is bounded below by 1.

Part (d), Step 1: give a rise and a fall for the sine sequence.

Let $s_n = \sin(n\pi/7)$. Put $c = \sin(\pi/7)$. Since $0 < \pi/7 < \pi$, the elementary sign property of sine gives $c > 0$. The sine values at three indices are

$$s_1 = c > 0, \qquad s_7 = \sin \pi = 0, \qquad s_{15} = \sin(2\pi + \pi/7) = c > 0.$$

An increasing sequence would satisfy $s_1 \le s_7$, contradicting $s_1 > 0 = s_7$. A decreasing sequence would satisfy $s_7 \ge s_{15}$, contradicting $s_7 = 0 < s_{15}$. Comparisons at nonadjacent indices are legitimate because monotonicity implies the corresponding comparison at any pair of ordered indices.

Part (d), Step 2: give global bounds.

The elementary trigonometric inequality $-1 \le \sin x \le 1$ is valid for every real x. Applying it at $x = n\pi/7$ gives $-1 \le s_n \le 1$ for every n.

Conclusion. The sequence $\sin(n\pi/7)$ is **bounded but neither increasing nor decreasing**.

Part (e), Step 1: rule out both monotonicity directions.

For $s_n = (-2)^n$, the first three terms are

$$s_1 = -2, \qquad s_2 = 4, \qquad s_3 = -8.$$

The rise from -2 to 4 rules out a decreasing sequence. The fall from 4 to -8 rules out an increasing sequence.

Part (e), Step 2: record an elementary power estimate.

We will use

$$2^j \geq j+1 \qquad (j \in \mathbb{N}).$$

For $j = 1$, this is $2 \geq 2$. If it holds for $j \geq 1$, then

$$2^{j+1} = 2 \cdot 2^j \geq 2(j+1) = 2j+2 \geq j+2,$$

since $(2j+2) - (j+2) = j \geq 0$. Induction proves the estimate.

Part (e), Step 3: show that both signs have arbitrarily large magnitude.

Let $R > 0$ be arbitrary. Choose $k \in \mathbb{N}$ with $k > R$ by the Archimedean property. At an even index,

$$s_{2k} = 2^{2k} \geq 2k+1 > R.$$

At the following odd index,

$$s_{2k+1} = -2^{2k+1} \leq -(2k+2) < -R.$$

Thus there are terms above R and below $-R$ for every positive R. In particular, no real number is an upper bound and no real number is a lower bound.

Conclusion. The sequence $(-2)^n$ is **neither increasing nor decreasing, and is unbounded both above and below**.

Part (f), Step 1: compare consecutive positive terms.

Here $s_n = n/3^n > 0$. Since we are dividing positive quantities, we can compute the consecutive-term ratio:

$$\begin{aligned} \frac{s_{n+1}}{s_n} &= \frac{n+1}{3^{n+1}} \frac{3^n}{n}, \\ &= \frac{n+1}{3n}. \end{aligned}$$

For $n \geq 1$, the denominator is positive and $3n-(n+1) = 2n-1 \geq 1 > 0$. Hence $0 < (n+1)/(3n) < 1$. Multiplying this strict upper inequality by $s_n > 0$ gives $s_{n+1} < s_n$.

Part (f), Step 2: bound the whole sequence.

Every term is positive. Repeated application of the decreasing inequality gives $s_n \leq s_1 = 1/3$. Therefore

$$0 < s_n \leq \frac{1}{3} \qquad (n \in \mathbb{N}).$$

The strict decrease also rules out an increasing sequence.

Conclusion. The sequence $n/3^n$ is **decreasing and bounded**, and is not increasing.

Summary of all six classifications.

Sequence	Increasing?	Decreasing?	Bounded?
$1/n$	No	Yes	Yes
$(-1)^n/n^2$	No	No	Yes
n^5	Yes	No	No
$\sin(n\pi/7)$	No	No	Yes
$(-2)^n$	No	No	No
$n/3^n$	No	Yes	Yes

10.2 Exercise 10.2: Convergence of a bounded decreasing sequence

What we must show. Prove the decreasing case of Theorem 10.2: every bounded decreasing real sequence converges.

Step 1: define the candidate limit and justify its existence.

Let (s_n) be bounded and decreasing, and let

$$S = \{s_n : n \in \mathbb{N}\}.$$

The set is nonempty because $s_1 \in S$, and it is bounded because the sequence is bounded. In particular, it is bounded below. The lower-bound form of completeness gives a real number

$$L = \inf S.$$

Being a lower bound means $L \leq s_n$ for every n.

Step 2: obtain one term close to the infimum.

Let $\varepsilon > 0$. The number $L + \varepsilon$ is strictly larger than the greatest lower bound L, so it cannot itself be a lower bound of S. Therefore there is an element of S below $L + \varepsilon$. By the definition of S, that element is s_N for some index $N \in \mathbb{N}$. Thus

$$L \leq s_N < L + \varepsilon.$$

Step 3: use decrease to control every subsequent term.

Take any integer $n > N$. Since the sequence is decreasing, $s_n \leq s_N$. Since L is a lower bound, $L \leq s_n$. Combining these facts with Step 2 gives

$$L \leq s_n \leq s_N < L + \varepsilon.$$

Subtract L throughout to get $0 \leq s_n - L < \varepsilon$. The difference $s_n - L$ is nonnegative, so

$$|s_n - L| = s_n - L < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. For every $\varepsilon > 0$ we obtained the threshold required by Definition 7.1. Hence $\boxed{s_n \longrightarrow \inf\{s_n : n \in \mathbb{N}\}}$. The proof uses the greatest-lower-bound property, without assuming the infimum is one of the terms.

10.3 Exercise 10.3: A finite bound for decimal approximations

What we must show. Let K be a nonnegative integer and let each digit d_j belong to $\{0, 1, \ldots, 9\}$. For

$$s_n = K + \sum_{j=1}^{n} \frac{d_j}{10^j},$$

prove $s_n < K + 1$ for every $n \in \mathbb{N}$.

Step 1: bound each digit contribution.

For every $j \geq 1$, $0 \leq d_j \leq 9$ and $10^j > 0$. Division by the positive denominator gives

$$0 \leq \frac{d_j}{10^j} \leq \frac{9}{10^j}.$$

Add these finitely many inequalities and then add K:

$$K \leq s_n \leq K + \sum_{j=1}^{n} \frac{9}{10^j}.$$

Step 2: evaluate the finite upper-bound sum by cancellation.

For an individual index j,

$$\frac{9}{10^j} = \frac{10-1}{10^j} = \frac{1}{10^{j-1}} - \frac{1}{10^j}.$$

Consequently

$$\begin{aligned} \sum_{j=1}^{n} \frac{9}{10^j} &= \left(1 - \frac{1}{10}\right) + \left(\frac{1}{10} - \frac{1}{10^2}\right) + \cdots + \left(\frac{1}{10^{n-1}} - \frac{1}{10^n}\right), \\ &= 1 - \frac{1}{10^n}. \end{aligned}$$

Every intermediate negative term cancels the identical positive term in the next parentheses. This is a finite algebraic identity; no limit or infinite sum has been used to derive it.

Step 3: preserve the strict final inequality.

Substitute the identity from Step 2 into Step 1:

$$K \leq s_n \leq K + 1 - \frac{1}{10^n}.$$

Since $10^n > 0$, its reciprocal is strictly positive. Subtracting that positive number makes $K+1-1/10^n$ strictly less than $K + 1$. Thus

$$s_n \leq K + 1 - \frac{1}{10^n} < K + 1.$$

Conclusion. Every finite decimal approximation satisfies $\boxed{s_n < K + 1}$. Moreover $s_{n+1} - s_n = d_{n+1}/10^{n+1} \geq 0$, so the approximations are increasing and bounded. Their convergence follows from Theorem 10.2 *after* the finite bound has been established.

10.4 Exercise 10.4: Why the monotone and Cauchy theorems require real completeness

What we must show. Explain why Theorems 10.2 and 10.11 would be false if the permitted terms and limits were restricted to $\mathbb{Q}$.

Plan. We construct a sequence of rational decimal truncations to $\sqrt{2}$. It is increasing, bounded by rational numbers, and Cauchy, but its only real limit is the irrational number $\sqrt{2}$. The construction itself uses finite sets and rational comparisons, rather than assuming that $\mathbb{Q}$ has the completeness property being tested.

Step 1: construct each truncation using integers only.

For each $n \in \mathbb{N}$, consider the finite set

$$E_n = \{k \in \mathbb{Z} : 0 \le k \le 2 \cdot 10^n,\ k^2 < 2 \cdot 10^{2n}\}.$$

It is nonempty because $0 \in E_n$. A finite nonempty set of integers has a largest member, so define

$$k_n = \max E_n, \qquad q_n = \frac{k_n}{10^n}.$$

The denominator is a positive integer, and the numerator is an integer. Hence every q_n is rational. The operations and comparisons used to define E_n involve only integers. In particular, this construction does not presuppose a supremum in $\mathbb{Q}$ for an infinite set.

Step 2: prove rational lower and upper bounds.

The integer 10^n belongs to E_n, since it is between 0 and $2 \cdot 10^n$ and its square 10^{2n} is strictly less than $2 \cdot 10^{2n}$. Thus $k_n \ge 10^n$. The integer $2 \cdot 10^n$ does not belong to E_n, since its square is $4 \cdot 10^{2n} > 2 \cdot 10^{2n}$. Therefore $k_n < 2 \cdot 10^n$. Dividing by $10^n > 0$ gives

$$1 \le q_n < 2.$$

Both bounds are rational, so the sequence is bounded even when all bounding numbers are required to lie in $\mathbb{Q}$.

Step 3: prove that the sequence is increasing.

Since $k_n^2 < 2 \cdot 10^{2n}$, multiplication by 100 gives

$$(10k_n)^2 < 2 \cdot 10^{2(n+1)}.$$

Also $0 \le 10k_n \le 2 \cdot 10^{n+1}$. Thus $10k_n \in E_{n+1}$. The maximality of k_{n+1} implies $10k_n \le k_{n+1}$. Dividing by $10^{n+1} > 0$ yields

$$q_n = \frac{10k_n}{10^{n+1}} \le \frac{k_{n+1}}{10^{n+1}} = q_{n+1}.$$

Equality may occur at some indices, which is permitted by the definition of increasing.

Step 4: give a rational upper approximation to the same square.

The integer $k_n + 1$ is still at most $2 \cdot 10^n$, because $k_n < 2 \cdot 10^n$ and both numbers are integers. If $(k_n + 1)^2 < 2 \cdot 10^{2n}$, then $k_n + 1$ would belong to E_n, contradicting maximality. Therefore

$$k_n^2 < 2 \cdot 10^{2n} \le (k_n + 1)^2.$$

Divide by $10^{2n} > 0$. The result is

$$q_n^2 < 2 \leq \left(q_n + \frac{1}{10^n}\right)^2.$$

Every number in this display is rational, so the comparisons remain meaningful entirely within $\mathbb{Q}$.

Step 5: verify the Cauchy condition through rational estimates.

For integers $m > n$, Step 3 gives $q_n \leq q_m$. The positive numbers q_m and $q_n + 10^{-n}$ satisfy

$$q_m^2 < 2 \leq (q_n + 10^{-n})^2.$$

Hence $q_m < q_n + 10^{-n}$. Indeed, the contrary inequality $q_m \geq q_n + 10^{-n} \geq 0$ would imply the contrary comparison between their squares. We conclude

$$0 \leq q_m - q_n < 10^{-n} \qquad (m > n).$$

We may use $10^j \geq j$ for every $j \in \mathbb{N}$: the case $j = 1$ is $10 \geq 1$; if $10^j \geq j$, then $10^{j+1} \geq 10j \geq j + 1$ because $j \geq 1$. This proves the estimate by induction, and taking reciprocals gives $10^{-j} \leq 1/j$.

Now let $\varepsilon > 0$ and choose $N = \max\{1, 1/\varepsilon\}$. Take any two integer indices $m, n > N$. If $m = n$, then $|q_m - q_n| = 0 < \varepsilon$. If $m > n$, the preceding estimate gives

$$|q_m - q_n| = q_m - q_n < 10^{-n} \leq \frac{1}{n} < \varepsilon.$$

If $n > m$, interchange their roles and use the symmetry of absolute value to obtain $|q_m - q_n| < 10^{-m} \leq 1/m < \varepsilon$. Thus the Cauchy estimate holds for every pair of indices above N. In particular, it holds for every positive rational tolerance, as a Cauchy condition formulated within $\mathbb{Q}$ requires.

Step 6: identify the only possible real limit.

In the ambient real numbers, let $\alpha = \sqrt{2}$. It is positive, $\alpha^2 = 2$, and it is irrational by Section 2. Step 4 and comparison of squares of positive numbers give

$$q_n < \alpha \leq q_n + 10^{-n}.$$

The upper equality is impossible because $q_n + 10^{-n}$ is rational and α is irrational. Therefore

$$0 < \alpha - q_n < 10^{-n} \leq \frac{1}{n}.$$

For any real $\varepsilon > 0$, the choice $N = \max\{1, 1/\varepsilon\}$ makes the right side strictly smaller than ε whenever $n > N$. Hence Definition 7.1 gives $q_n \to \alpha$ as a sequence in $\mathbb{R}$.

Step 7: rule out every rational limit.

A limit in $\mathbb{Q}$ would have to be some $r \in \mathbb{Q}$. Restricting the definition to positive rational tolerances makes no difference: given any real tolerance $\varepsilon > 0$, the Archimedean property gives a positive rational $1/j < \varepsilon$, and the estimate for that smaller rational tolerance implies the estimate for ε. Thus a limit in $\mathbb{Q}$ would also be the same limit in $\mathbb{R}$.

Because r is rational and α is irrational, they are distinct. Put $d = |\alpha - r| > 0$. If q_n converged to both numbers, there would be thresholds N_1, N_2 such that, for $n > \max\{N_1, N_2\}$,

$$|q_n - \alpha| < \frac{d}{3}, \qquad |q_n - r| < \frac{d}{3}.$$

The triangle inequality would then give

$$d = |\alpha - r| \leq |\alpha - q_n| + |q_n - r| < \frac{2d}{3} < d,$$

a contradiction. Thus the rational sequence has no rational limit.

Conclusion. The same rational sequence is bounded and increasing, so it contradicts the version of Theorem 10.2 with limits restricted to $\mathbb{Q}$. It is also Cauchy, so it contradicts the corresponding version of Theorem 10.11. Both failures arise because the required limit $\sqrt{2}$ is missing from $\mathbb{Q}$, not because the sequence is unbounded or fails its Cauchy estimates.

10.5 Exercise 10.5: An unbounded decreasing sequence tends to negative infinity

What we must show. Prove Theorem 10.4(ii): if a real sequence (s_n) is decreasing and unbounded, then $s_n \to -\infty$.

Step 1: determine which direction must be unbounded.

Since the sequence is decreasing, every term satisfies $s_n \leq s_1$. Thus s_1 is an upper bound of its set of terms. If the sequence also had a real lower bound, it would be bounded above and below, contrary to the hypothesis that it is unbounded. Therefore the sequence is unbounded below.

Step 2: find a term below an arbitrary target.

Let $M < 0$ be any real target. Since the sequence is unbounded below, M cannot be a lower bound for all its terms. Hence some index $N \in \mathbb{N}$ satisfies

$$s_N < M.$$

Step 3: show that every later term remains below that target.

For every integer $n > N$, decrease gives $s_n \leq s_N$. Thus

$$s_n \leq s_N < M \qquad (n > N).$$

We have proved that for each $M < 0$ there exists a threshold N such that every later term is less than M.

Conclusion. This is the definition of divergence to negative infinity, so $\boxed{s_n \longrightarrow -\infty}$. The choice of s_N would in fact work for any real target M, not just negative ones.

10.6 Exercise 10.6: Small successive differences: one sufficient rate and one insufficient rate

What we must show. Prove (a) that $|s_{n+1} - s_n| < 2^{-n}$ for every natural n implies (s_n) is Cauchy and convergent; and (b) decide whether the same conclusion follows from the weaker estimate $|s_{n+1} - s_n| < 1/n$.

Part (a), Step 1: compare arbitrary terms by telescoping.

Fix integers $m > n$. Then

$$s_m - s_n = \sum_{k=n}^{m-1} (s_{k+1} - s_k).$$

To see the cancellation, expand the right side as $(s_{n+1} - s_n) + (s_{n+2} - s_{n+1}) + \cdots + (s_m - s_{m-1})$; every intermediate term occurs once with each sign. Applying the finite triangle inequality gives

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| < \sum_{k=n}^{m-1} 2^{-k}.$$

The final inequality is strict because the finite sum is nonempty and every one of its summands is strictly less than the corresponding 2^{-k}.

Part (a), Step 2: bound this finite geometric sum.

For an individual k, $2^{-k} = 2^{1-k} - 2^{-k}$. Substituting that identity into the sum makes the consecutive terms telescope:

$$\begin{aligned}\sum_{k=n}^{m-1} 2^{-k} &= (2^{1-n} - 2^{-n}) + (2^{-n} - 2^{-n-1}) + \cdots + (2^{2-m} - 2^{1-m}),\\ &= 2^{1-n} - 2^{1-m},\\ &< 2^{1-n}.\end{aligned}$$

The last inequality holds because $2^{1-m} > 0$. We have therefore shown

$$|s_m - s_n| < 2^{1-n} \qquad (m > n).$$

Part (a), Step 3: choose one threshold for all pairs.

Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ such that

$$N > \max\left\{1, \frac{2}{\varepsilon}\right\}.$$

The elementary estimate $2^N \geq N + 1$ was proved in Exercise 10.1(e). It gives

$$2^{1-N} = \frac{2}{2^N} \leq \frac{2}{N+1} < \frac{2}{N} < \varepsilon.$$

Now let $m, n > N$ be arbitrary integer indices. If $m = n$, their terms have distance $0 < \varepsilon$. If $m > n$, Steps 1–2 give

$$|s_m - s_n| < 2^{1-n} \leq 2^{1-N} < \varepsilon.$$

If $n > m$, interchange the indices and use $|s_m - s_n| = |s_n - s_m|$ to get the same conclusion with 2^{1-m} in place of 2^{1-n}.

Conclusion. For every $\varepsilon > 0$ all pairs of terms after the chosen threshold have distance less than ε. Thus (s_n) is Cauchy by Definition 10.8, and Theorem 10.11 for real sequences implies that it converges to a real number.

Part (b), Step 1: construct a sequence satisfying the strict hypothesis.

Define the harmonic partial sums

$$H_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

The consecutive difference is exactly

$$H_{n+1} - H_n = \frac{1}{n+1}.$$

Both sides are positive, and $n + 1 > n > 0$, so

$$|H_{n+1} - H_n| = \frac{1}{n+1} < \frac{1}{n} \qquad (n \in \mathbb{N}).$$

The required inequality is strict at every index, including $n = 1$.

Part (b), Step 2: show directly that the sequence is not Cauchy.

Fix the positive tolerance $\varepsilon_0 = 1/2$. Given any proposed threshold N, choose an integer $n > \max\{N, 1\}$ and put $m = 2n$. Both indices are greater than N. The difference of the terms is

$$H_{2n} - H_n = \sum_{k=n+1}^{2n} \frac{1}{k}.$$

There are exactly n summands. For each index in this sum, $k \leq 2n$, so $1/k \geq 1/(2n)$. Adding the n inequalities gives

$$|H_{2n} - H_n| = H_{2n} - H_n \geq n\frac{1}{2n} = \frac{1}{2} = \varepsilon_0.$$

Thus, no matter how N is chosen, some pair of indices $m, n > N$ violates the required strict Cauchy estimate.

Conclusion. The answer to part (b) is **no**. The sequence (H_n) satisfies every strict bound $|H_{n+1} - H_n| < 1/n$, but it is not Cauchy. Since every convergent sequence is Cauchy by Lemma 10.9, it does not converge.

10.7 Exercise 10.7: Approaching an unattained supremum by points of the set

What we must show. Let $S \subseteq \mathbb{R}$ be bounded and nonempty, and suppose $\sup S \notin S$. Construct a sequence (s_n) with $s_n \in S$ for every n and $s_n \to \sup S$.

Step 1: use the least-upper-bound property at each precision.

Completeness gives the real number $M = \sup S$. For each $n \in \mathbb{N}$, we have $1/n > 0$, and consequently $M - 1/n < M$. The smaller number $M - 1/n$ cannot be an upper bound of S, because M is its least upper bound. Therefore there is a point $s_n \in S$ satisfying $M - 1/n < s_n$. Since M is an upper bound, also $s_n \leq M$. Equality is ruled out by $M \notin S$. Thus, choosing such a point for each n, we obtain

$$M - \frac{1}{n} < s_n < M \qquad (n \in \mathbb{N}).$$

Step 2: translate the order estimate into an error estimate.

Subtracting s_n and rearranging the left inequality gives $M - s_n < 1/n$. The right inequality gives $M - s_n > 0$, so

$$|s_n - M| = M - s_n < \frac{1}{n}.$$

This bound holds at every index; the actual choices of the points s_n need not produce a monotone sequence.

Step 3: verify convergence with an explicit threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, $n > 1/\varepsilon$ gives $1/n < \varepsilon$, and therefore

$$|s_n - M| < \frac{1}{n} < \varepsilon.$$

Conclusion. The constructed sequence consists entirely of points of S and satisfies $\boxed{s_n \longrightarrow \sup S}$. The supremum itself is excluded from the set, so each approximation is strictly below its target.

10.8 Exercise 10.8: Averages of an increasing positive sequence

What we must show. Let (s_n) be increasing with $s_n > 0$ for every n, and let

$$\sigma_n = \frac{1}{n}\sum_{k=1}^{n} s_k.$$

Prove that the average sequence (σ_n) is increasing in the sense of Definition 10.1.

Step 1: compare each average with the last term it averages.

Fix $n \in \mathbb{N}$. Since (s_n) is increasing, every k with $1 \le k \le n$ satisfies $s_k \le s_n$. Add these n inequalities:

$$s_1 + s_2 + \cdots + s_n \le \underbrace{s_n + s_n + \cdots + s_n}_{n \text{ terms}} = ns_n.$$

Divide by the positive integer n to obtain $\sigma_n \le s_n$. Since $s_n \le s_{n+1}$, transitivity gives

$$\sigma_n \le s_n \le s_{n+1}.$$

In particular, $s_{n+1} - \sigma_n \ge 0$.

Step 2: express the next average in terms of the previous one.

The definition of σ_n gives $s_1 + \cdots + s_n = n\sigma_n$. Therefore

$$\begin{aligned}\sigma_{n+1} &= \frac{s_1 + \cdots + s_n + s_{n+1}}{n+1},\\ &= \frac{n\sigma_n + s_{n+1}}{n+1}.\end{aligned}$$

Step 3: calculate the exact difference between consecutive averages.

Put the terms over the same positive denominator $n+1$:

$$\begin{aligned}\sigma_{n+1}-\sigma_n &= \frac{n\sigma_n+s_{n+1}}{n+1}-\frac{(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-(n+1)\sigma_n}{n+1},\\ &= \frac{n\sigma_n+s_{n+1}-n\sigma_n-\sigma_n}{n+1},\\ &= \frac{s_{n+1}-\sigma_n}{n+1}.\end{aligned}$$

The numerator is nonnegative by Step 1, and the denominator $n+1$ is positive. Hence $\sigma_{n+1}-\sigma_n \geq 0$, or $\sigma_n \leq \sigma_{n+1}$.

Conclusion. This comparison holds for every $n \in \mathbb{N}$, proving $\boxed{(\sigma_n) \text{ is increasing}}$. Equality can occur: for a constant positive sequence, all the averages are equal. Thus the weak meaning of "increasing" is the one required by the hypotheses and conclusion.

10.9 Exercise 10.9: A decreasing recurrence with a squared term

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{n}{n+1}s_n^2 \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove that a finite limit exists, and prove that its value is 0.

Part (a): substitute the indices one at a time.

For $n = 1$, the recurrence gives

$$s_2 = \frac{1}{1+1}s_1^2 = \frac{1}{2}\cdot 1^2 = \frac{1}{2}.$$

For $n = 2$,

$$s_3 = \frac{2}{2+1}s_2^2 = \frac{2}{3}\left(\frac{1}{2}\right)^2 = \frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}.$$

For $n = 3$,

$$s_4 = \frac{3}{3+1}s_3^2 = \frac{3}{4}\left(\frac{1}{6}\right)^2 = \frac{3}{4}\cdot\frac{1}{36} = \frac{3}{144} = \frac{1}{48}.$$

Conclusion. $\boxed{s_2 = \frac{1}{2}, \quad s_3 = \frac{1}{6}, \quad s_4 = \frac{1}{48}}$.

Part (b), Step 1: prove positivity and an upper bound by induction.

We claim $0 < s_n \leq 1$ for every $n \in \mathbb{N}$.

The initial case is $s_1 = 1$, so $0 < s_1 \leq 1$ holds. For the induction step, assume $0 < s_n \leq 1$. Since $n \geq 1$, the coefficient satisfies

$$0 < \frac{n}{n+1} < 1.$$

Also $0 < s_n^2 \leq 1$. Multiplying these positive quantities gives

$$0 < s_{n+1} = \frac{n}{n+1}s_n^2 < 1.$$

This proves the claim by induction.

Part (b), Step 2: compare consecutive terms.

The inequality $s_n \le 1$, multiplied by the positive number s_n, gives $s_n^2 \le s_n$. The coefficient $n/(n+1)$ is strictly below 1, and $s_n^2 > 0$, so

$$s_{n+1} = \frac{n}{n+1} s_n^2 < s_n^2 \le s_n.$$

Hence the sequence is strictly decreasing. It is bounded below by 0 and above by 1.

Part (b), Step 3: apply monotone convergence before manipulating a limit.

Theorem 10.2 for bounded monotone sequences now applies. There is a real number L with $s_n \to L$. The inequalities $s_n > 0$ imply $L \ge 0$ by Exercise 8.9(a). Since the sequence is decreasing and $s_2 = 1/2$,

$$0 < s_n \le \frac{1}{2} \qquad (n \ge 2).$$

Exercise 8.9 also passes this eventual upper bound to the limit. Therefore

$$0 \le L \le \frac{1}{2}.$$

Conclusion. A finite limit exists, and it belongs to $[0, 1/2]$. Its existence has been proved independently of any equation for its value.

Part (c), Step 1: justify the shifted limit and the coefficient limit.

If $s_n \to L$, then $s_{n+1} \to L$ as well. Indeed, for any $\varepsilon > 0$ choose a threshold N with $m > N \Rightarrow |s_m - L| < \varepsilon$. For $n > N$, we have $n + 1 > N$, so the same estimate holds with $m = n + 1$.

Next put $c_n = n/(n+1)$. Its error from 1 is

$$|c_n - 1| = \left| \frac{n - (n+1)}{n+1} \right| = \frac{1}{n+1}.$$

For any $\varepsilon > 0$, if $n > 1/\varepsilon$, then $n + 1 > 1/\varepsilon$, so $1/(n+1) < \varepsilon$. Thus $c_n \to 1$.

Part (c), Step 2: apply the product limit law to the established limits.

Because $s_n \to L$ and $c_n \to 1$, the product laws from Section 9 give

$$\lim_{n\to\infty} c_n s_n^2 = 1 \cdot L \cdot L = L^2.$$

The recurrence says that this is the same sequence as s_{n+1}, whose limit is L. Uniqueness of a finite limit gives

$$L = L^2.$$

Rearranging and factoring gives

$$L^2 - L = 0, \qquad L(L-1) = 0.$$

The zero-product property therefore gives $L = 0$ or $L = 1$.

Part (c), Step 3: use the tail bound to remove the extraneous candidate.

Part (b) proved $L \le 1/2$, so $L = 1$ is impossible. The only remaining candidate is $L = 0$.

Conclusion. $\boxed{\lim_{n\to\infty} s_n = 0}$.

An additional direct error bound.

The same recurrence supplies a quantitative check. Since $s_n^2 \le s_n$,

$$s_{n+1} \le \frac{n}{n+1} s_n.$$

Starting with $s_1 = 1 = 1/1$, induction gives $s_n \le 1/n$: if $s_n \le 1/n$, then

$$s_{n+1} \le \frac{n}{n+1} \cdot \frac{1}{n} = \frac{1}{n+1}.$$

Consequently

$$|s_n - 0| = s_n \le \frac{1}{n} < \varepsilon \qquad \left(n > \frac{1}{\varepsilon}\right).$$

This also verifies the zero limit directly from Definition 7.1.

10.10 Exercise 10.10: A linear recurrence approaching one half

What we must show. For

$$s_1 = 1, \qquad s_{n+1} = \frac{s_n + 1}{3} \quad (n \in \mathbb{N}),$$

compute s_2, s_3, s_4, prove $s_n > 1/2$ by induction, prove that the sequence is decreasing, and identify its limit.

Part (a): calculate the first three new terms.

Using the recurrence successively gives

$$s_2 = \frac{1+1}{3} = \frac{2}{3},$$

$$s_3 = \frac{\frac{2}{3}+1}{3} = \frac{\frac{2}{3}+\frac{3}{3}}{3} = \frac{5/3}{3} = \frac{5}{9},$$

and

$$s_4 = \frac{\frac{5}{9}+1}{3} = \frac{\frac{5}{9}+\frac{9}{9}}{3} = \frac{14/9}{3} = \frac{14}{27}.$$

Conclusion. $\boxed{s_2 = \frac{2}{3}, \quad s_3 = \frac{5}{9}, \quad s_4 = \frac{14}{27}}$.

Part (b), Step 1: verify the initial case of the proposed lower bound.

The proposition for induction is P_n: $s_n > 1/2$. For $n = 1$, it says $1 > 1/2$, which is true.

Part (b), Step 2: prove the induction implication.

Assume $s_n > 1/2$. Adding 1 to both sides gives $s_n + 1 > 3/2$. Dividing by the positive number 3 preserves the strict inequality:

$$s_{n+1} = \frac{s_n + 1}{3} > \frac{3/2}{3} = \frac{1}{2}.$$

Thus P_n implies P_{n+1}. Induction proves the bound for every positive integer n.

Conclusion. $\boxed{s_n > \frac{1}{2} \text{ for every } n \in \mathbb{N}}$.

Part (c): subtract consecutive terms and determine the sign.

Use a common denominator:

$$s_{n+1} - s_n = \frac{s_n + 1}{3} - s_n = \frac{s_n + 1 - 3s_n}{3} = \frac{1 - 2s_n}{3}.$$

Part (b) gives $2s_n > 1$, so the numerator is negative. The denominator 3 is positive. Hence $s_{n+1} - s_n < 0$, which means $s_{n+1} < s_n$ for every n.

Conclusion. The sequence is strictly decreasing.

Part (d), Step 1: establish convergence from bounds and monotonicity.

The terms satisfy $1/2 < s_n \leq s_1 = 1$, and part (c) proves that the sequence is decreasing. Theorem 10.2 therefore gives a finite limit L. Exercise 8.9 implies

$$\frac{1}{2} \leq L \leq 1.$$

Part (d), Step 2: solve the limit equation after convergence is known.

The shifted sequence s_{n+1} has the same limit L, by the threshold argument given in Exercise 10.9(c). The addition and scalar-multiplication laws from Section 9 give

$$\lim_{n\to\infty} \frac{s_n + 1}{3} = \frac{L + 1}{3}.$$

The recurrence identifies these two sequences, so their limits agree:

$$L = \frac{L + 1}{3}.$$

Multiplying by 3 and collecting terms gives

$$3L = L + 1, \qquad 2L = 1, \qquad L = \frac{1}{2}.$$

Conclusion. $\boxed{\lim_{n\to\infty} s_n = \frac{1}{2}}$. Every term is strictly greater than $1/2$, but the limit can equal that lower bound.

10.11 Exercise 10.11: An infinite-product recurrence

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) prove that a finite limit exists and (b) investigate its value. Part (b) asks what the limit appears to be. We will distinguish a numerical guess from a proof; an optional elementary-calculus argument will then verify the exact value.

Part (a), Step 1: bound the coefficient and prove positivity.

For every $n \geq 1$, we have $4n^2 \geq 4$, so

$$0 < \frac{1}{4n^2} \leq \frac{1}{4}.$$

Subtracting from 1 gives

$$\frac{3}{4} \leq 1 - \frac{1}{4n^2} < 1.$$

In particular, every recurrence coefficient is strictly positive. The initial value is $t_1 = 1 > 0$. If $t_n > 0$, the recurrence is a product of two positive numbers and gives $t_{n+1} > 0$. Induction shows $t_n > 0$ for all n.

Part (a), Step 2: prove monotonicity and apply the theorem.

Since $1 - 1/(4n^2) < 1$ and $t_n > 0$,

$$t_{n+1} = \left(1 - \frac{1}{4n^2}\right) t_n < t_n.$$

Thus (t_n) is strictly decreasing, bounded below by 0 and above by its first term 1. Theorem 10.2 gives a finite real limit L. The bound on the terms implies $0 \leq L \leq 1$ by Exercise 8.9.

Conclusion. The limit of (t_n) exists as a real number. This conclusion uses only positivity, monotonicity, and boundedness.

Part (b), Step 1: compute initial values and a numerical table.

The first terms are

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{16}\right) \frac{3}{4} = \frac{15}{16} \cdot \frac{3}{4} = \frac{45}{64},$$

and

$$t_4 = \left(1 - \frac{1}{36}\right) \frac{45}{64} = \frac{35}{36} \cdot \frac{45}{64} = \frac{1575}{2304} = \frac{175}{256}.$$

Continuing the recurrence produces the following rounded values:

n	t_n
1	1.000000000000
2	0.750000000000
3	0.703125000000
4	0.683593750000
10	0.653587392299
100	0.638221323951
1000	0.636779026837

The numbers suggest a limit near 0.63662. The familiar constant $2/\pi$ has the decimal value $0.636619772368\ldots$, which suggests the guess

$$L = \frac{2}{\pi}.$$

The numerical agreement motivates this guess; finitely many numerical calculations by themselves do not prove the exact identification.

Part (b), Step 2: explain why a direct limit equation gives no value.

The coefficient $1 - 1/(4n^2)$ tends to 1. Taking limits in the recurrence, after convergence has been proved, only yields

$$L = 1 \cdot L.$$

Every real L satisfies this identity, so it does not determine the limit. Further information is needed to justify the guess $2/\pi$.

Optional verification of the guess: the elementary-calculus input.

The remaining argument uses integration by parts and the order property of definite integrals. Those are additional calculus tools; they were not needed for the existence proof in part (a). We derive the required product formula here rather than assume Wallis's product.

For an integer $m \geq 0$, define

$$I_m = \int_0^{\pi/2} \sin^m x \, dx,$$

where $\sin^0 x$ means the constant function 1. The integrands are continuous and are positive in the interior when $m \geq 1$, so all I_m are positive. Direct integration gives

$$I_0 = \frac{\pi}{2}, \qquad I_1 = 1.$$

Optional verification, Step 1: derive the integral recurrence.

For $m \geq 2$, write $I_m = \int_0^{\pi/2} \sin^{m-1} x \sin x \, dx$. In integration by parts take

$$u = \sin^{m-1} x, \quad dv = \sin x \, dx, \quad du = (m-1)\sin^{m-2} x \cos x \, dx, \quad v = -\cos x.$$

The boundary term $[-\sin^{m-1} x \cos x]_0^{\pi/2}$ is zero: $\cos(\pi/2) = 0$, and at 0 the factor $\sin^{m-1} 0$ is zero because $m - 1 \geq 1$. Consequently

$$\begin{aligned} I_m &= (m-1) \int_0^{\pi/2} \sin^{m-2} x \cos^2 x \, dx \\ &= (m-1) \int_0^{\pi/2} \sin^{m-2} x (1 - \sin^2 x) \, dx \\ &= (m-1)(I_{m-2} - I_m). \end{aligned}$$

Adding $(m-1)I_m$ to both sides and dividing by $m > 0$ gives

$$I_m = \frac{m-1}{m} I_{m-2}.$$

Optional verification, Step 2: express the even and odd integrals.

Repeated use of the recurrence, ending at I_0 or I_1, gives for every integer $N \geq 1$,

$$I_{2N} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{2k-1}{2k}, \qquad I_{2N+1} = \prod_{k=1}^{N} \frac{2k}{2k+1}.$$

For example, the even recurrence successively contributes the factors $(2N-1)/(2N)$, $(2N-3)/(2N-2)$, down to $1/2$. The odd recurrence contributes $(2N)/(2N+1)$ down to $2/3$. These are exactly the displayed finite products.

Unrolling the recurrence for t_n gives

$$t_{N+1} = \prod_{k=1}^{N} \left(1 - \frac{1}{4k^2}\right) = \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2}.$$

The initial empty product for t_1 is 1, and each new recurrence step multiplies by the next factor, which justifies this finite-product identity. Dividing the formulas for the positive integrals gives

$$\frac{I_{2N}}{I_{2N+1}} = \frac{\pi}{2} \prod_{k=1}^{N} \frac{(2k-1)(2k+1)}{(2k)^2} = \frac{\pi}{2}\, t_{N+1}.$$

Optional verification, Step 3: bound the integral ratio.

On $[0, \pi/2]$, the quantity $\sin x$ lies in $[0, 1]$. Multiplication by a number in $[0, 1]$ cannot increase a nonnegative power, so

$$\sin^{2N+1} x \le \sin^{2N} x \le \sin^{2N-1} x.$$

Integrating preserves these inequalities:

$$I_{2N+1} \le I_{2N} \le I_{2N-1}.$$

Divide by $I_{2N+1} > 0$. The integral recurrence with $m = 2N + 1$ gives $I_{2N+1} = (2N)/(2N + 1)\, I_{2N-1}$, so

$$1 \le \frac{I_{2N}}{I_{2N+1}} \le \frac{I_{2N-1}}{I_{2N+1}} = \frac{2N+1}{2N} = 1 + \frac{1}{2N}.$$

Substituting $I_{2N}/I_{2N+1} = (\pi/2)t_{N+1}$ and multiplying by $2/\pi > 0$ yields

$$0 \le t_{N+1} - \frac{2}{\pi} \le \frac{1}{\pi N}.$$

For any $\varepsilon > 0$, a positive integer index $n > 1 + 1/(\pi\varepsilon)$ has $N = n - 1 > 1/(\pi\varepsilon)$. Therefore

$$\left| t_n - \frac{2}{\pi} \right| \le \frac{1}{\pi(n-1)} < \varepsilon.$$

This proves the conjectured limit by Definition 7.1.

Conclusion. Part (a) proves existence using Section 10 alone. For part (b), the computed terms suggest $2/\pi$. The explicitly marked additional calculus argument verifies that

$$\boxed{\lim_{n\to\infty} t_n = \frac{2}{\pi}.}$$

No product formula was taken on trust; the finite-product identity and its limiting estimate were derived.

10.12 Exercise 10.12: A product recurrence with an exact formula

What we must show. For

$$t_1 = 1, \qquad t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) t_n \quad (n \in \mathbb{N}),$$

(a) establish existence of the limit, (b) guess its value, (c) prove $t_n = (n+1)/(2n)$ by induction, and (d) use the formula to verify the limit.

Part (a), Step 1: prove positivity and monotonicity.

For every $n \geq 1$, we have $n+1 \geq 2$, so

$$0 < \frac{1}{(n+1)^2} \leq \frac{1}{4}, \qquad \frac{3}{4} \leq 1 - \frac{1}{(n+1)^2} < 1.$$

The initial term is positive. If $t_n > 0$, multiplying it by this positive coefficient gives $t_{n+1} > 0$. Induction proves positivity for every term.

Because the coefficient is strictly less than 1 and $t_n > 0$, we also have $t_{n+1} < t_n$ for every n. Thus the sequence is strictly decreasing, bounded below by 0 and above by $t_1 = 1$.

Part (a), Step 2: apply bounded monotone convergence.

Theorem 10.2 now gives a finite limit L. Exercise 8.9 implies $0 \leq L \leq 1$.

Conclusion. The limit exists as a real number. Its value has not been assumed in establishing existence.

Part (b): calculate early terms to formulate a guess.

Successive substitutions give

$$t_2 = \left(1 - \frac{1}{4}\right) 1 = \frac{3}{4},$$

$$t_3 = \left(1 - \frac{1}{9}\right) \frac{3}{4} = \frac{8}{9} \cdot \frac{3}{4} = \frac{2}{3},$$

$$t_4 = \left(1 - \frac{1}{16}\right) \frac{2}{3} = \frac{15}{16} \cdot \frac{2}{3} = \frac{5}{8},$$

and

$$t_5 = \left(1 - \frac{1}{25}\right) \frac{5}{8} = \frac{24}{25} \cdot \frac{5}{8} = \frac{3}{5}.$$

The values

$$1, \quad \frac{3}{4}, \quad \frac{2}{3}, \quad \frac{5}{8}, \quad \frac{3}{5}$$

fit the proposed pattern $t_n = (n+1)/(2n)$. Since this expression equals $1/2 + 1/(2n)$, it suggests the limit $1/2$. At this stage, the pattern and the limit are guesses to be proved.

Part (c), Step 1: state the induction proposition and verify its base.

Let P_n be the assertion

$$t_n = \frac{n+1}{2n}.$$

For $n = 1$, the right side is $(1+1)/(2 \cdot 1) = 1$, which equals t_1. Thus P_1 holds.

Part (c), Step 2: substitute the induction hypothesis.

Assume P_n for an arbitrary positive integer n. The recurrence gives

$$t_{n+1} = \left(1 - \frac{1}{(n+1)^2}\right) \frac{n+1}{2n}$$

$$
\begin{aligned}
&= \frac{(n+1)^2 - 1}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n^2 + 2n}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n(n+2)}{(n+1)^2} \frac{n+1}{2n} \\
&= \frac{n+2}{2(n+1)} \\
&= \frac{(n+1)+1}{2(n+1)}.
\end{aligned}
$$

The cancellation in the penultimate line is legitimate because $n > 0$ and $n+1 > 0$, so neither cancelled factor is zero. The final expression is precisely the formula in P_{n+1}.

Conclusion. Mathematical induction proves

$$\boxed{t_n = \frac{n+1}{2n} \quad \text{for every } n \in \mathbb{N}.}$$

Part (d), Step 1: separate the constant term from the error.

The proved formula gives the identity

$$t_n = \frac{n+1}{2n} = \frac{n}{2n} + \frac{1}{2n} = \frac{1}{2} + \frac{1}{2n}.$$

Since $1/n \to 0$, the limit laws from Section 9 give $t_n \to 1/2$.

Part (d), Step 2: verify the conclusion directly with a threshold.

For any $\varepsilon > 0$, choose $N = 1/(2\varepsilon)$. Every positive integer $n > N$ satisfies $1/(2n) < \varepsilon$, and

$$\left| t_n - \frac{1}{2} \right| = \left| \frac{1}{2n} \right| = \frac{1}{2n} < \varepsilon.$$

This is the definition of convergence to $1/2$.

Conclusion. The guess in part (b) is now proved:

$$\boxed{\lim_{n \to \infty} t_n = \frac{1}{2}.}$$

The exact formula also gives the error $t_n - 1/2 = 1/(2n)$ for every positive integer index.